VIT Vidyalankar Institute of Technology

End Semester Examination (MU)

Evaluation of University Question Paper Solution Prepared by Faculty

Subject (Write in full) : Computer Graphics **(Regular / KT)**

Exam : May 20 25 **/ Nov. 20** 26 **Exam Date :** 18/05/25 **Q. Paper Code :** PCIT03

Department : I.T **Year : FE/SE/TE/BE/ME/MMS Semester :** 5 **Scheme :** CBSGS/CBCS

Handwritten Solution Prepared by : Debarati Ghosal Debarati

Name of the Subject Cluster : Virtual Graphics

Name of the Cluster Mentor / Assessor : Dr Shubhajit Jawade

1st Assessment :

Question No.	Marks Obtained						Total
1	(a)	(b)	(c)	(d)	(e)	(f)	
2	(a)	(b)	(c)	(d)	(e)	(f)	
3	(a)	(b)	(c)	(d)	(e)	(f)	
4	(a)	(b)	(c)	(d)	(e)	(f)	
5	(a)	(b)	(c)	(d)	(e)	(f)	
6	(a)	(b)	(c)	(d)	(e)	(f)	
Total (Out of marks)							

Signature of Assessor / Cluster Mentor : _____

2nd Assessment :

Question No.	Marks Obtained						Total
1	(a)	(b)	(c)	(d)	(e)	(f)	
2	(a)	(b)	(c)	(d)	(e)	(f)	
3	(a)	(b)	(c)	(d)	(e)	(f)	
4	(a)	(b)	(c)	(d)	(e)	(f)	
5	(a)	(b)	(c)	(d)	(e)	(f)	
6	(a)	(b)	(c)	(d)	(e)	(f)	
Total (Out of marks)							

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Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks	Question No.	START WRITING HERE

Define Pixel Resolution, Raster Scan

8 a) in games & virtual worlds. other equally popular area includes,

- Virtual Museums
- Galleries
- Theatre
- Virtual theme parks
- Discovery centers

5. Fashion :- Fashion is not something that immediatly springs to mind when thinking about VR but nevertheless, it is used by the fashion industry in a variety of ways.

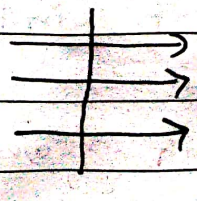
This includes :-

- VR s/w for building virtual fashion stores
- 3D avatars
- Fashion show in second life
- 3D fashion portfolio.

b) Parallel & perspective projection :-

Parallel projection :-

- Orthographic project^{n}
  - projectors orthogonal to the view plane.

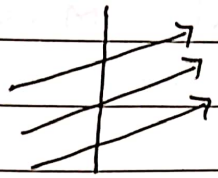


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Oblique projection

- projectors not orthogonal to the view plane.

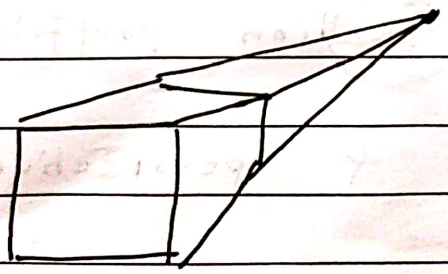


Perspective projection :-

2 most charact. features

- smaller as their distance from the observer increases.
- the size of an object's dimensions along the line of sight are relatively shorter than dimension across the line of sight.

A drawing has one-pt perspective when it contains only one vanishing pt on the horizon line.



c) Need for homogeneous matrix represent^{n} :-

- one of the many purposes of using homogeneous coordinates is to capture the concept of infinity.

In the Euclidian coordinate system, infinity is something that does

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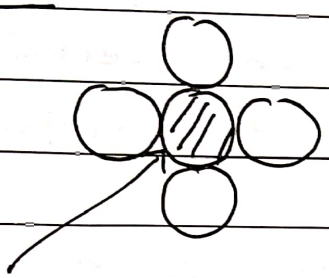
Space for Marks	Question No.	START WRITING HERE
		Explain Aliasing, Antialiasing, Pres Part.
	Q1b)	Without the use of this, it would be diff. to design certain classes of very useful curves & surfaces in CG & CAD.
		let us consider 2 real no's a & w, & compute the value of a/w. let us hold the value of a fixed & vary the value of w. As w getting smaller, a/w getting larger. If w → 0, a/w approaches to ∞.
		Thus to capture the concept of ∞, we use 2 no's a & w to represent a & v, v = a/w. if w is not 0, the value is exactly a/w otherwise, we identify the value with (a, 0). Therefore the concept of infinity can be represented with a number path (a, w) or as a quotient a/w.
	d)	Boundary fill :-
		It is used to fill polygon but the boundary colors must be same.
		filling start from the pt. that lies inside the polygon called as seed pt.
		2 approaches are used :-

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Re-Assessor

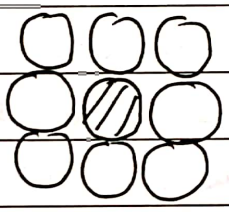
e for ks | Question No. | START WRITING HERE

4 - connected :



Whenever seed pt filled with color, it's adjacent 4 pixels also filled with same color.

8 - connected :



Whenever seed pt filled with color, it's adjacent 4 pixels & 4 diagonal pixels also filled with same color.

```
bfill ( x, y, bc, fc)
{
  if ( getpixel (x,y) != bc && getpixel (x,y) != fc)
     {
          putpixel (x, y, fc)
          bfill (x-1, y, bc, fc)
          bfill (x+1, y, bc, fc)
          bfill (x, y-1, bc, fc)
          bfill (x, y+1, bc, fc)
     }
}
```

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		Re-Assessor	

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8	c)

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Polygon set of vertex coordinates

Flood fill :- apply outside test using Even Odd fu

This polygon filling algo. is used even if the boundary colors of polygon are different.

This algo. also uses 4- connected & 8- connected approach.

```
ffill (x, y, oc, fc)
{
    if ( getpixel (x,y) == oc)
        {
            putpixel (x, y, fc)
            ffill ( x-1, y , oc, fc)
            ffill ( x+1, y , oc, fc )
            ffill ( x , y+1, oc, fc )
            ffill ( x , y-1 , oc , fc)
        }
}
```

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ace for Marks | Question No.

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what is difference between local & global give importance of illumination

Q 2

a) Bresenham's line drawing algo.

1. Start
2. Enter the coordinates of 2 end pts of line $(x1, y1)$ & $(x2, y2)$
3. Find the value of dx & dy.

$dx = abs(x2 - x1)$

$dy = abs(y2 - y1)$

4. Find the value of xinc & yinc.

i.e. if $(x1 < x2)$

xinc = 1

else

xinc = -1

if $(y1 < y2)$

yinc = 1

else

yinc = -1

5. Plot the 1st pt. of a line.

i.e. $x = x1$

$y = y1$

putpixel (x, y, color)

6. plot the remaining pts

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		Re-Assessor	

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i-e.

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```
if (dx >= dy)   // abs(slope) <= 1
{
    p = 2dy - dx
    while( x != x2)
    {
        x = x + xinc
        if (p <= 0)
            p= p + 2dy
        else
            { y = y + yinc
              p = p + 2 (dy - dx)
            }
        putpixel (x,y, color)
    }
}
else
{
    p = 2dx - dy
    while( y != y2)
    {
        y = y + yinc
        if (p <= 0)
            p = p + 2dx
        else
            {
                x = x + xinc
                p = p + 2 (dx - dy)
            }
        putpixel (x,y, color)
    }
}
```

& 7. Stop.

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Different from DDA

- DDA involves floating pt. arithmetic whereas Bresenham's includes integer arithmetic.

- Bresenham's algo is fast & accurate.

- It contain less no. of operations.

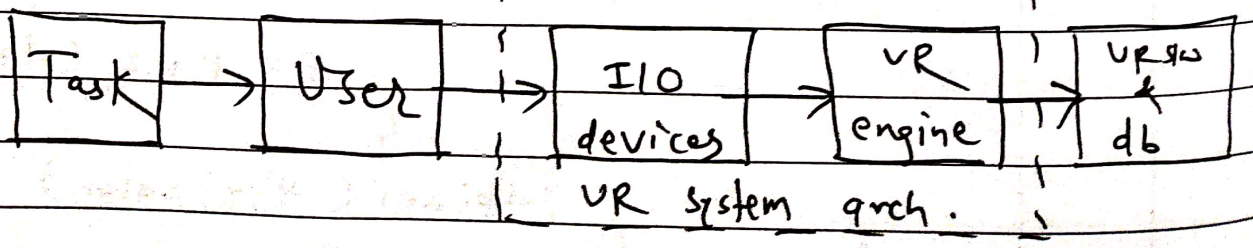
Q.26) Virtual Reality :Explain visible surface detection in computer graphics

It is an interactive computer generated experience taking place within a simulated environment.

It incorporates mainly auditory & visual feedback, but may also allow other types of sensory feedback like haptic.

The immersive env. can be similar to the real world or it can be fantastical.

Components of VR :-



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Task :- There are various types of tasks that are performed on VR like display of static & dynamic imgs, animatns, image warping, etc.

User :-

From multiple task, user select one of the task thrugh GUI. These GUI must be def. properly, so that any user can interact with it.

I/O devices :-

Using this, users can enter the i/p's for selected task & after task completn, o/p's are displayed on devices like video monitor, PC graphics display, etc.

VR engine :-

It is a special purpose computer architecture which is design to match high I/O & computatn demands of real time VR simuln.

VR S/w & db :-

There are var. types of VR S/w's like geometry texture, Java 3D, WTK which are design to help VR applns.

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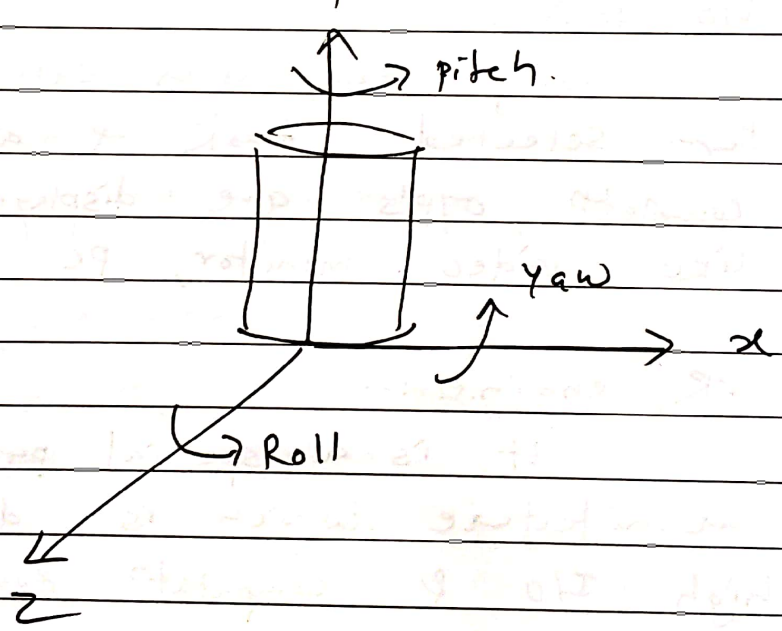
Q.2a) I/O devices of VR

Q26 Identify difference between translation, rotation

3D Tracker :- Most applicatns like Scaling navigatn, missile tracking, robotics, bio-mechanics, CAD requires the knowledge of real time posn & orientn of moving objects.

- Posn & orientn can be described in terms of dof.

- A moving object in 3D space has atleast 6 dof, 3 for translⁿ & 3 for rotatn.



Trackball :- It is used for navigatn & manipuln. It consists of sensing cylinder at the center that measures 3 forces & 3 torques applied user's hand.

The central pt has 6 LED & on the

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moving external part of a Cylinder are 6 photo sensors.

Gesture Interface :-

These are sensing gloves that have embeded sensors that measures the posⁿ of fingers.

Pinch Glove :-

It is the only sensing glove that makes calibratⁿ unnecessary.

The gloves incarporates electrodes in the form of conducting filter that patch at finger tips back of finger & on the palm.

5DT data glove

Whenever there is any incremental changes, pinch glove cannot be used.

5DT data glove is a sensing glove having one sensor for finger & the filled sensor to me wrist orientⁿ.

Personal Graphic display

HMD can be used to capture imges which are in front of eyes shutter

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distance. older HMD has very low resolut^n but it is improved in HMD

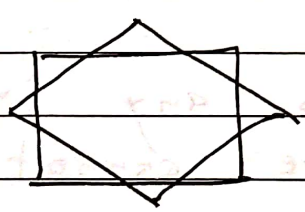
what is difference between DDA & Bresenham line drawing algorithm (9,18) (14,22)

Q.3 Sutherland Hodgeman polygon clipping

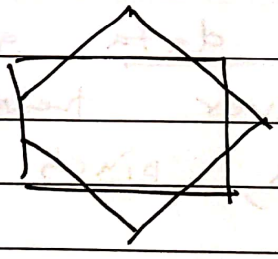
Vertices are processed against 4 clipping boundaries.

Initially, vertices are check against left clipping boundary to produce new set of vertices.

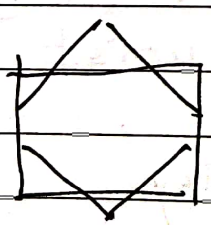
These vertices are check against right boundary then bottom & finally top boundary.



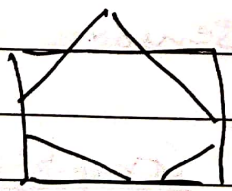
initial



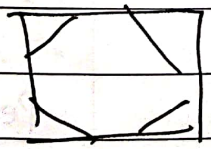
left Clipping



Right clipping



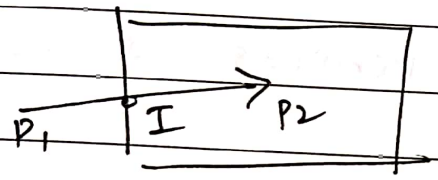
Bottom Clipping



top Clipping

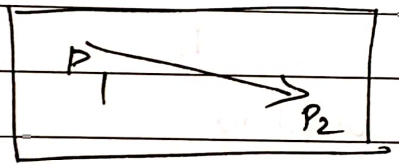
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Rules :— 1) If 1st vertex is outside & 2nd vertex is inside the clipping window then save intersect^n pt & 2nd vertex.



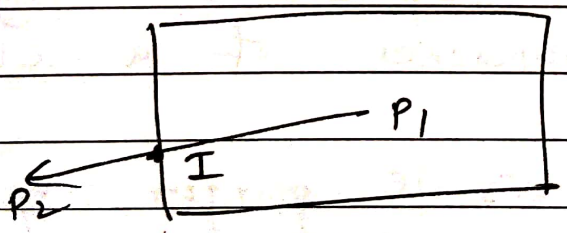
Save I, P2

2) If both the vertices lies inside the clipping window then save only 2nd vertex.



Save P2

3) If 1st vertex lies inside & 2nd vertex lies outside then save only intersect^n pt-



Save I

4) If both vertices lies outside the window then don't save any vertex



Save nothing.

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Q 3b) Give detailed explanation of Midpoint circle drawing algorithm, give application.

Algorithm :-

1) Start

2) Enter no. of vertices & their coordinates of a polygon.

3) Enter window coordinates (x_{wmin}, y_{wmin}) & (x_{wmax}, y_{wmax})

draw clipping window.

4) Initially consider left clipping boundary of a window.

5) Process polygon vertices against considered boundary to produce new set of vertices by using above 4 rules.

6) Repeat step 5 for right, bottom & top boundaries of a clipping window.

7) draw visible polygon by connecting the vertices which are extracted from o/p vertex list.

8) display visible polygon

9) stop.

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Apply 2D transformations in order

Q.49) Curve: A line or outline which gradually deviates from being straight for some or all of it's length.

Bezier curve can be constructed by accepting n+1 control pts.

Any control pt. is given by

$$P(u) = \sum_{k=0}^{n} P_k \cdot B_{n,k}(u).$$

where

$$P_k = (x_k, y_k, z_k)$$

$B_{n,k}(u)$: Blending fn which is given by,

$$= {}^{n}C_k \, u^k (1-u)^{n-k}.$$

Let no. of control pts = 4

$\therefore n+1 = 4$

$\therefore n = 3$

$$B_{3,0}(u) = {}^{3}C_0 \, u^0 (1-u)^3 = (1-u)^3$$

$$B_{3,1}(u) = {}^{3}C_1 \, u^1 (1-u)^2 = 3u(1-u)^2$$

$$B_{3,2}(u) = {}^{3}C_2 \, u^2 (1-u)^1 = 3u^2(1-u)$$

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$B_{3,3}(u) = {}^3C_3\, u^3 (1-u)^0 = u^3$

By using this, blending fn, Bezier curve can be constructed as,

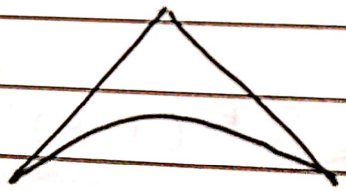
$P(u) = P_0 \cdot B_{n,0}(u) + P_1 B_{n,1}(u) + P_2 B_{3,2}(u)$

$+ P_3 B_{n,3}(u)$

Properties

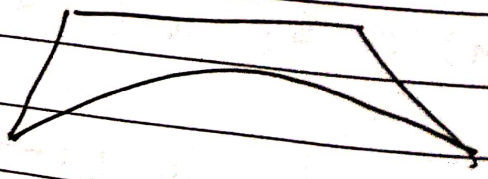
- Bezier curve always pass through under the convex hull

e.g.



- It always pass through 1st & last control pts.

e.g.



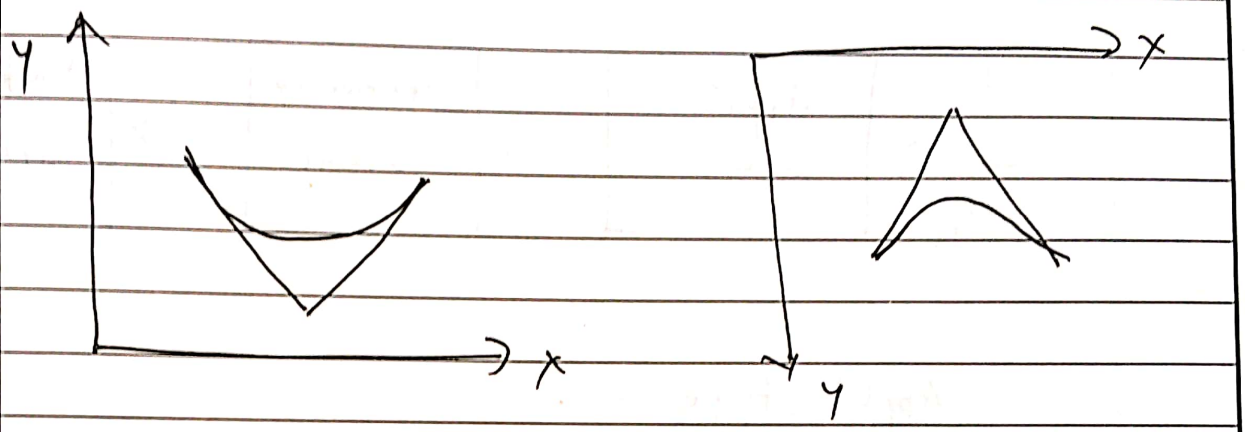
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- It is independent of axis orientatn.

e.g.



- It is independent of rotn & translⁿ.

- Shape of Bezier curve may change if control pt. coordinates changes.

- It possess variatn diminishing property

- It follows the shape of defining polygon

- The order of Bezier curve is always one less than the no. of control pts.

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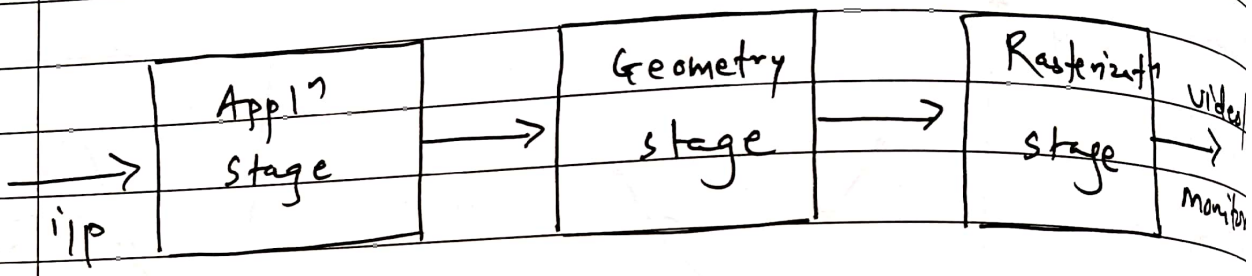
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Homogeneous coordinate system

Graphics Rendering pipeline :-

It consist of 3 stages



Appl^n stage :-

- It is implemented in S/w.
  - Runs on several CPU's
- Read the world geometry database & the user's i/p

Basic operations
- take care of user's i/p
- collision detection algo.

- O/p : 3D geometric primitives

Geometry stage :-

- It is implemented in s/w or h/w
  - Run on the geometry engines
- Basic operations

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- Model transformatns
- Light computatn
- screen projectn
- clipping
- screen mapping

Output : 2D geometric primitives

Rasterizatn stage

- Implemented in H/W.
- Convert the vertex infn o/p into pixel infn needed by the video display

- Basic opns
  - scan conversion
  - z-buffering
  - Anti aliasing
  - Texture mapping.

o/p : pixel values.

Bottleneck

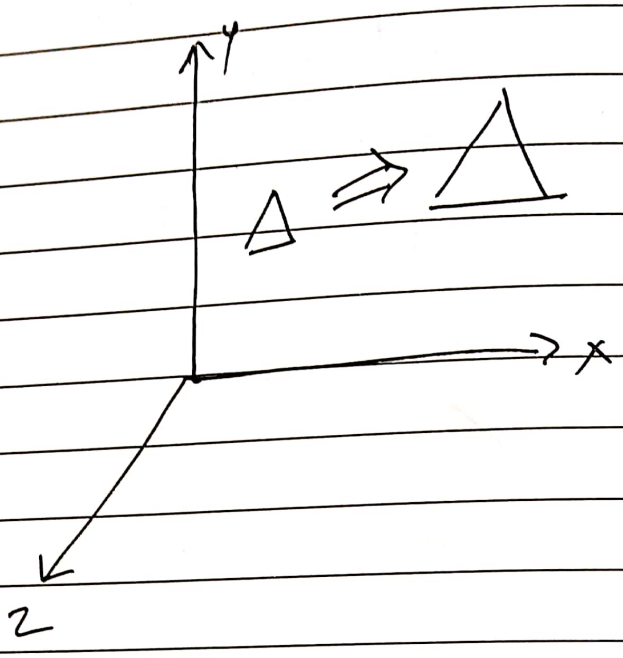
- one of the 3 stages will be slowest

- Cpu limited
  - In the Apptn stage
- Transform limited
  - In the geometry stage.

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$x' = S_x \cdot x$

$y' = S_y \cdot y$

$z' = S_z \cdot z$

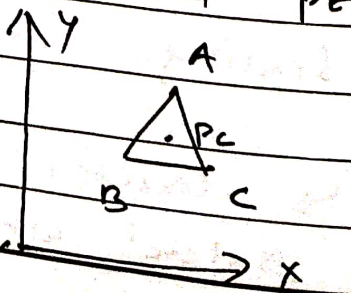
Scaling matrix is,

$$S = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

New coordinates can be calc. as,

$$[A' \; B' \; C'] = S \cdot [A \quad B \quad C]$$

2) Scaling wrt. fixed pt.



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